

EC824B

Post Selection Inference

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LASSO

- Model with $X \in \mathbb{R}^{N \times K}$ where $K \gg N$

$$Y = X\beta + \varepsilon,$$

- LASSO: ℓ_1 -norm penalised regression

$$\operatorname{argmin}_{\beta \in \mathbb{R}^K} \frac{1}{2N} \|Y - X\beta\|^2 \text{ s.t. } \|\beta\|_1 \leq R_1 \equiv \operatorname{argmin}_{\beta \in \mathbb{R}^K} \frac{1}{2N} \|Y - X\beta\|^2 + \lambda_1 \|\beta\|_1$$

- Sought out λ_1 s.t. $MSE(\hat{\beta}) < MSE(\hat{\beta}_{OLS})$
- How to do inference on post LASSO regression
- This lecture follows closely: [Hastie, Tibshirani and Wainwright (2015)] CH 6, [Lee et al. (2016)]

- 1 Post selection inference
- 2 Post selection inference - Forward stepwise regression
- 3 Post selection inference - LASSO
- 4 Data splitting and carving

Post selection inference

Can we simply run OLS and use our usual standard errors on the post selection subset of variables?

- NO: these standard errors fail to account for the fact we have selected our model, such that inference depends on the model we select
- What we need is *inference conditional* on the model selected

If the LASSO selected model is $\hat{M}$ (using notation popular in this literature) then we want the probability of rejecting the test T conditional on a *random* hypothesis, $\hat{H}_0$:

$$\text{Prob}\left(T \text{ rejected} | \hat{M}, \hat{H}_0\right) \leq \alpha$$

where α is our desired Type I error

- Our hypothesis, $\hat{H}_0$, is now a random variable!
- Ignoring the selection event in the condition leads to over-optimistic Type I errors!

Post selection inference

How do we get inference conditional on an event? Let's focus on

- $Y \sim N(\mu, \sigma^2 \mathbb{I}_{N \times N})$, $X_1, X_2, \dots, X_K$ and potential models,

$$Y = X_{\mathcal{M}} \beta^M + \varepsilon$$

where $\mathcal{M}$ simply indexes the variables included in model M

- We would like to do inference on β^M accounting for the fact we have selected M .
Confidence intervals (conditional vs. classic):

$$\text{Prob}(\beta_j^{\widehat{M}} \in C_j^{\widehat{M}} | \widehat{M} = M) \quad \text{vs.} \quad \left\{ \text{Prob}(\beta_j^{\widehat{M}} \in C_j^{\widehat{M}}) \text{ only for } j \in \widehat{\mathcal{M}} \right\}$$

In fact, the events inside the probability statement on the right hand side are not well defined for $j \notin \widehat{\mathcal{M}}$, which is why we focus on $j \in \widehat{\mathcal{M}}$

Post selection inference (intuition)

Why should we worry about

$$Prob(\beta_j^{\widehat{M}} \in C_j^{\widehat{M}} | \widehat{M} = M) \quad \text{vs.} \quad \left\{ Prob(\beta_j^M \in C_j^{\widehat{M}}) \text{ only for } j \in \widehat{\mathcal{M}} \right\}$$

The RHS probability conditions on the event that Y conditional on X is free to vary over any variation admitted by the model. However, the requirement $\widehat{\mathcal{M}} = \mathcal{M}$ restricts this:

$$\mathcal{R} := \{ \{Y|X\} \text{ admissible set that selects } \widehat{M} \} \quad \text{vs.} \quad \mathcal{U} := \{ \{Y|X\} \text{ admitted by } \widehat{M} \}$$

If we erroneously assume $\{Y|X\}$ is bounded by the model $\widehat{M}$'s full support, we may be over optimistic

Post selection inference (intuition)

Say $\mathcal{R}$ restricts a coefficient to having absolute value greater than some value. E.g. $|\beta_j^{\widehat{M}}| > \lambda$.

$$Prob(reject | \widehat{\beta} \in \mathbb{R}, H_0) = \alpha_{\text{Naïve}} \quad \text{vs.} \quad Prob(reject | |\widehat{\beta}| > \lambda, H_0) = \alpha_{\text{PostSelection}}$$

If $\widehat{se}(\widehat{\beta}) = 1$ and $H_0 : \beta = 0$,

$$\begin{aligned} \alpha_{\text{Naïve}} = 0.05 &= Prob(|\widehat{\beta}| > 1.96 | \widehat{\beta} \in \mathbb{R}, \beta = 0) \\ &\leq Prob(|\widehat{\beta}| > 1.96 | |\widehat{\beta}| > \lambda, \beta = 0) \\ &= \alpha_{\text{PostSelection}} \\ &\quad (\text{which correspond to 1.96 crit. val. under restricted model}) \end{aligned}$$

- Hence, the Type 1 error can be understated under the Naïve hypothesis test
- Confidence intervals can be very understated (too tight) under classical inference

Post selection inference - Example (File drawer effect)

- Sample N independent measurements of $Y_i \sim N(\mu_i, 1)$
- Only “large” effects are selected:

$$\hat{I} = \{i : |Y_i| > 1\}$$

- Want to test $H_{0,i} : \mu_i = 0$ for each i at $\alpha = 0.05$
- Question: Why is the rejection region $|Y_i| > 1.96$ invalid?
- First: For given $H_{0,i}$, $\mathbb{P}(|Y_i| > 1.96) = 0.05 \rightarrow$ seemingly nothing invalid yet
- However: For *collection of selected* $H_{0,i}$, the type 1 error might be higher

Post selection inference - Example (File drawer effect)

- **Definition of error rate conditional on selection:**

$$\frac{\# \text{ false rejections}}{\# \text{ true nulls selected}} = \frac{\frac{1}{N_0} \sum_{i: H_{0,i} \text{ true}} 1\{i \in \hat{I}, \text{reject } H_{0,i}\}}{\frac{1}{N_0} \sum_{i: H_{0,i} \text{ true}} 1\{i \in \hat{I}\}} \rightarrow \mathbb{P}_{H_{0,i}}(\text{reject} \mid i \in \hat{I})$$

where $N_0 \xrightarrow{N \rightarrow \infty} \infty$ is the number of true nulls. Then the nominal test is $\Phi(-1.96)/\Phi(-1) \approx 0.16$

- Here, we can adjust to control at $\alpha = 0.05$ by adjusting our critical value:

$$\Phi(-c)/\phi(-1) = 0.05 \Leftarrow c = 2.41$$

hence, we must be more strict!

- **Guiding principle:**

“ The answer must be valid, given that the question was asked. ”

i.e. selected hypothesis usually validate themselves, if not adjusted

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Post selection inference - Forward stepwise regression

The restriction

$$\mathcal{R} := \{\{Y|X\} \text{ admissible set that selects } \hat{M}\}$$

is vague and difficult to define for any generating process. Luckily, the selection event implied by LASSO can be boiled down to linear restrictions on Y . First, something easier:

- Forward stepwise regression: sequentially add regressors that optimise objective error
- Classic inference:

2k - 1. Pick regressor with largest inner product and add to regression, $e_0 := Y - \bar{Y}$,

$$\operatorname{argmin}_j |\langle X_j, e_{k-1} \rangle| \quad \text{where } e_{k-1} \text{ is error term with first } k-1 \text{ FSR regressors}$$

2k. Perform OLS with usual standard errors for each k (usual rejection region, etc.)

Post selection inference - Forward stepwise regression

In steps $2k$ we fail to account for the selection of the first (strong) regressors

- The FSR selection event can be described by a set of linear restrictions on Y ,

$$\{AY \leq b\}$$

- In the case of the FSR, set $b = 0$, and let the rows of A be the different restrictions that imply the sequential absolute inner product dominance for each X_k
- For linear predictors, $\eta'Y$, the restrictions $\{AY \leq b\}$ imply bounds on the set of possible values $\eta'Y$ can take, thus restricting the conditioning set we should be doing inference on

Post selection inference - Forward stepwise regression

Let's take a simple example to understand $\{AY \leq b\}$:

- $X_1, X_2, \|X_k\| = 1$ for each $k = 1, 2$ and $|\langle X_1, Y \rangle| > |\langle X_2, Y \rangle|$
- When FSR selects X_1 in the first iteration,

$$A = \begin{bmatrix} -s_1 X_1' + X_2' \\ -s_1 X_1' - X_2' \end{bmatrix} \quad \text{where} \quad s_k := \text{sign}(\langle X_k, Y \rangle)$$

For $\{AY \leq b\}$ for $b = 0$ conveniently forms the restrictions

$$-s_1 X_1' Y \leq X_2' Y \leq s_1 X_1' Y$$

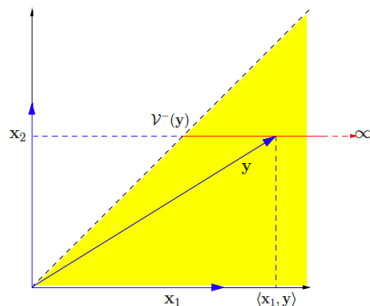
which is exactly $|\langle X_1, Y \rangle| \geq |\langle X_2, Y \rangle|$

We would like to use this to form bounds on $\eta' Y$ (which in this example in the first step of FSR is simply $\langle X_1, Y \rangle$ from $\|X_k\| = 1$)

Post selection inference - Forward stepwise regression

$$\begin{aligned}
 -s_1 X_1' Y \leq X_2' Y \leq s_1 X_1' Y &\equiv -s_1 \eta' Y \leq X_2' Y \leq s_1 \eta' Y \\
 &\equiv \begin{cases} \max\{-X_2' Y, X_2' Y\} \leq \eta' Y & \text{if } s_1 = 1 \\ \min\{-X_2' Y, X_2' Y\} \geq \eta' Y & \text{if } s_1 = -1 \end{cases}
 \end{aligned}$$

if we focus on the positive-positive quadrant for $\langle X_1, Y \rangle, \langle X_2, Y \rangle$, with $X_1 \perp X_2$



$\eta' Y \geq X_2' Y$ because we focus on $+/+$ quadrant
for fixed X_1, X_2 , this bounds $\eta' Y \in [X_2' Y, \infty)$

Universal bounds for $\eta' Y$ labelled $\mathcal{V}^-(Y), \mathcal{V}^+(Y)$

Take max(min) of the lower(upper) bounds implied
by $\{AY \leq b\}$

Post selection inference - Forward stepwise regression

Focus on $s_1 = s_2 = 1$ such that our support is $\eta'Y \in [X_2'Y, \infty)$. Then we need to bound the domain of our normal CDF for inference. For generic bound $[c,d]$:

$$F_{\mu, \sigma^2}^{c,d}(x) = \frac{\Phi((x - \mu)/\sigma) - \Phi((c - \mu)/\sigma)}{\Phi((d - \mu)/\sigma) - \Phi((c - \mu)/\sigma)}$$

is a bounded normal distribution correction. Hence, for our application,

$$F_{\mu, \sigma^2}^{X_2'Y, \infty}(x) = \frac{\Phi((x - \mu)/\sigma) - \Phi((X_2'Y - \mu)/\sigma)}{1 - \Phi((X_2'Y - \mu)/\sigma)}$$

Remember, $F_X(X) \sim U(0,1)$ such that, for $Y \sim N(\mu, \sigma^2 \mathbb{I}_N)$,

$$F_{\eta'\mu, \sigma^2 \|\eta\|^2}^{X_2'Y, \infty}(\eta'Y) | \{AY \leq b\} \sim U(0,1)$$

which now has the correct domain!

Post selection inference - Forward stepwise regression

What about inference on $\beta = \eta' \mu$ (population statistic of interest)?

- Define $P(\beta) = F_{\beta, \sigma^2 \|\eta\|^2}^{X_2' Y, \infty}(\eta' Y) | \{AY \leq b\}$
- Then lower CI at α is the value of β that achieves $1 - P(\beta) = \alpha/2$
- Upper CI at α is the value of β that achieves $P(\beta) = \alpha/2$

Post selection inference - Polyhedra lemma

We have seen a very specific example to motivate inference on a restricted conditioning set implied by $\{AY \leq b\}$. In general settings, this can imply many more restrictions on how a linear estimator of $\eta'\mu$ may vary. [Lee et al. (2016); Tibshirani et al. (2016)] show

$$\{AY \leq b\} = \{\mathcal{V}^-(Y) \leq \eta'Y \leq \mathcal{V}^+(Y), \mathcal{V}^0(Y) \geq 0\}$$

where,

$$\alpha := \frac{A\eta}{\|\eta\|^2},$$

$$\mathcal{V}^-(Y) = \max_{j:\alpha_j < 0} \frac{b_j - (AY)_j + \alpha_j \eta'Y}{\alpha_j}$$

$$\mathcal{V}^+(Y) = \min_{j:\alpha_j > 0} \frac{b_j - (AY)_j + \alpha_j \eta'Y}{\alpha_j}$$

$$\mathcal{V}^0(Y) = \min_{j:\alpha_j = 0} (b_j - (AY)_j)$$

This complicated equivalence statement is called the *polyhedra lemma*

Post selection inference - Polyhedra lemma

The restriction $\{AY \leq b\}$, and polyhedra lemma,

$$\{AY \leq b\} = \{\mathcal{V}^-(Y) \leq \eta'Y \leq \mathcal{V}^+(Y), \mathcal{V}^0(Y) \geq 0\}$$

tell us that as we continue to add constraints, our bounds on $\eta'\mu$ can only get tighter. This implies our adjusted CDF, $c = \mathcal{V}^-(Y)$, $d = \mathcal{V}^+(Y)$, $\tilde{\mu} = \eta'\mu$, $\tilde{\sigma}^2 = \sigma^2\|\eta\|^2$

$$F_{\tilde{\mu}, \tilde{\sigma}^2\|\eta\|^2}^{c,d}(x) = \frac{\Phi((x - \tilde{\mu})/\tilde{\sigma}) - \Phi((c - \tilde{\mu})/\tilde{\sigma})}{\Phi((d - \tilde{\mu})/\tilde{\sigma}) - \Phi((c - \tilde{\mu})/\tilde{\sigma})}$$

considers a tighter and tighter domain ($[c, d]$ gets narrower)

- It's not clear if $F(x)$ is increasing/decreasing as $[c, d]$ gets narrower - depends on x
- But we do expect it to change with the bounds
- Type I errors are usually understated in practice if these bounds are ignored

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Post selection inference - LASSO

- Remember our solution (normalise X s.t. $X'X = \mathbb{I}_K$):

$$\hat{\beta}_k = \begin{cases} \hat{\beta}_{OLS,k} - \lambda_1 \text{sign}(\beta_{OLS,k}) & \text{if } |\hat{\beta}_{OLS,k}| \geq \lambda_1 \\ 0 & \text{otherwise} \end{cases}$$

so the selection criteria for post-LASSO estimation is $|\hat{\beta}_{OLS,k}| \geq \lambda_1$

We can also find the selection region for LASSO. Let us use a simple example again:

- $X_1, X_2, \|X_k\| = 1$ for each $k = 1, 2$ and $X_1 \perp X_2$, let LASSO pick only $\tilde{k} = 1$
- From our LASSO solution we know:
 - $|\hat{\beta}_{OLS,1}| \geq \lambda_1$
 - $|\hat{\beta}_{OLS,2}| \leq \lambda_1$
 - $\|X_k\| = 1 \Rightarrow |\hat{\beta}_{OLS,k}| = s_k X_k' Y$ where $s_k = \text{sign}(X_k' Y)$
- This is adequate information to form our linear restrictions on Y !

Post selection inference - LASSO

This selection implies,

$$AY \leq b \Rightarrow \begin{bmatrix} -s_1 X_1' \\ s_2 X_2' \end{bmatrix} Y \leq \begin{bmatrix} -\lambda_1 \\ \lambda_1 \end{bmatrix}$$

So we get our

$$A = \begin{bmatrix} -s_1 X_1' \\ s_2 X_2' \end{bmatrix}, \quad \text{and} \quad b = \begin{bmatrix} -\lambda_1 \\ \lambda_1 \end{bmatrix}$$

Draw plot of selection region for Y projected onto X_1, X_2

Post selection inference - LASSO

Our object $\eta'Y$ is the same, because we are doing post-LASSO

$$\eta'Y = X_1'Y$$

So $\eta' = X_1'$

- We have our ingredients now to form

$$\{AY \leq b\} = \{\mathcal{V}^-(Y) \leq \eta'Y \leq \mathcal{V}^+(Y), \mathcal{V}^0(Y) \geq 0\}$$

(we needed A , b , and an expression of our estimator in linear form, $\eta'Y$)

Post selection inference - LASSO

$$\alpha = \frac{A\eta}{\|\eta\|^2} = \frac{AX_1}{\|X_1\|^2} = \begin{bmatrix} -s_1 X_1' X_1 \\ s_2 X_2' X_1 \end{bmatrix} = \begin{bmatrix} -s_1 \\ 0 \end{bmatrix}$$

$$\begin{cases} \mathcal{V}^- = \frac{b_1 - (AY)_1 + (-s_1 \eta' Y)}{-s_1} & \text{if } s_1 = 1 \text{ s.t. } \alpha_1 < 0 \\ \mathcal{V}^+ = \frac{b_1 - (AY)_1 + (-s_1 \eta' Y)}{-s_1} & \text{if } s_1 = -1 \text{ s.t. } \alpha_1 > 0 \\ \mathcal{V}^0 = b_2 - (AY)_2 & \text{because } \alpha_2 = 0 \end{cases}$$

$$= \begin{cases} \mathcal{V}^- = \lambda_1 & \text{if } s_1 = 1 \text{ s.t. } \alpha_1 < 0 \\ \mathcal{V}^+ = -\lambda_1 & \text{if } s_1 = -1 \text{ s.t. } \alpha_1 > 0 \\ \mathcal{V}^0 = \lambda_1 - (AY)_2 & \text{because } \alpha_2 = 0 \end{cases}$$

Post selection inference - LASSO

Hypothesis test:

$$H_0 : \beta = \eta' \mu = 0 \quad \text{vs.} \quad H_1 : \beta = \eta' \mu \neq 0$$

We can use, for $H_0 : \eta' \mu = 0$

$$F_{\eta' \mu, \sigma^2 \|\eta\|^2}^{\lambda_1, \infty}(\eta' Y) | \{AY \leq b\} \sim U(0, 1), \quad \text{or} \quad F_{\eta' \mu, \sigma^2 \|\eta\|^2}^{-\infty, -\lambda_1}(\eta' Y) | \{AY \leq b\} \sim U(0, 1)$$

for $s_1 = 1$, respectively $s_1 = -1$. Key here is that we know the distribution of our test statistic

- Very simple rejection rule: reject H_0 when $\eta' Y$ in the upper or lower $\alpha/2$ tails of the truncated distribution
- E.g. reject H_0 if: $\min \{1 - F_{0, \sigma^2 \|\eta\|^2}^{\lambda_1, \infty}(\eta' Y), F_{0, \sigma^2 \|\eta\|^2}^{-\infty, -\lambda_1}(\eta' Y)\} < \alpha/2$

Post selection inference - LASSO

Let's compare to classic inference

- $Y \sim N(0, \mathbb{I}_{N \times N})$, $\|\eta\|^2 = 1$
- By classical hypothesis,

$$F_{\eta'\mu, \sigma^2\|\eta\|^2}^{-\infty, \infty}(\eta'Y) = F_{0,1}^{-\infty, \infty}(\eta'Y) = \Phi(\eta'Y) \sim U(0, 1)$$

- By truncation hypothesis,

$$F_{0,1}^{\lambda_1, \infty}(\eta'Y) = \frac{\Phi(\eta'Y) - \Phi(\lambda_1)}{1 - \Phi(\lambda_1)} \sim U(0, 1), \quad F_{0,1}^{\infty, -\lambda_1}(\eta'Y) = \frac{\Phi(\eta'Y)}{\Phi(-\lambda_1)} \sim U(0, 1)$$

- Question: for what values of $\eta'Y$ is classical more likely to overreject compared to truncated?

Post selection inference

We learnt from the above exercises that model selection has non-trivial implications on how we should think about *conditional* confidence intervals,

$$Prob(\beta_j^{\widehat{M}} \in C_j^{\widehat{M}} | \widehat{M} = M) \quad \text{vs.} \quad \left\{ Prob(\beta_j^{\widehat{M}} \in C_j^{\widehat{M}}) \text{ only for } j \in \widehat{\mathcal{M}} \right\}$$

- The setting has assumed $Y \sim N(\mu, \sigma^2 \mathbb{I}_{N \times N})$
- However, usually in econometrics we don't like to impose distributional assumptions
- E.g. for OLS we can rely on the CLT to get normally distributed $\widehat{\beta}$ estimators, *without* imposing normal errors
- Can the CLT be applied in the model selection context? This seems conceptually difficult as a lot of the analysis relies on $Y \sim N(\mu, \sigma^2 \mathbb{I}_{N \times N})$

Post selection inference

The takeaway message remains:

- Be wary of estimation analysis that has utilised extensive model selection
- p -values can be very inaccurate with model selection - this comes from the spuriously tight confidence intervals
- The issue is that data is used twice:
 1. First, to select the model,
 2. Second, to estimate the model and do inferencethis is why our hypothesis is now random
- A simple workararound is to split the sample - use one part to select the model and the other to estimate
 - Beware: this can have power issues if N is not very, very large

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Data splitting

Before we made the argument that because our null hypothesis is a function of our sample, it is a random variable. Hence, loosely,

$$\text{Prob}\left(T \text{ rejected} | \hat{M}, \hat{H}_0\right) \neq \text{Prob}\left(T \text{ rejected}\right)$$

What if we can make the selection event $\{\hat{M}, \hat{H}_0\}$ independent of our test? That is, what if we can make our inference stage blind to our selection stage? Then, it is natural that

$$\text{Prob}\left(T \text{ rejected} | \hat{M}, \hat{H}_0\right) = \text{Prob}\left(T \text{ rejected}\right)$$

because the test is independent of the conditioning $|\hat{M}, \hat{H}_0$

- This is the notion of classical inference: that any information that goes into formulating a model and hypothesis is completely independent of the data used for inference
- Philosophically, this is very hard to justify!

Data splitting

One simple solution is to make sure the data we use for selection is independent of the data we use for post selection estimation and inference:

- Y_i is independently distributed over i
- Split $i = \{1, \dots, N\}$ into two blocks, $\{S_1, S_2\}$
- Select your model using S_1 : $\hat{M}(S_1)$, and formulate $\hat{H}_0(S_1)$
- Estimate model and test statistic using S_2 : $\hat{\beta}(S_2)$, $\hat{se}(S_2)$, and $\hat{T}(S_2)$
- Now the formation of our hypothesis and model selection is independent from our estimated test statistic - can use classic inference techniques with S_2

But, we lose data

1. Lose data do to model selection: less confidence in our choice of model
2. Lose a lot of power in the inference stage (and not reject false nulls) from this decrease in effective sample size

Data splitting - Example

“DGP” (in high-dimensional settings, people usually avoid DGPs or true models):

- Sample size N
- $X_i \in \mathbb{R}^K, \varepsilon_i \sim i.i.d.$ (or even just independently) over i
- Y_i is generated by a linear model of a subset of X_i , but we don't know which subset:

$$Y_i = X'_{i,\mathcal{M}} \beta^M + \varepsilon_i$$

Method:

- Split data into $\{S_1; S_2\} = \{1, \dots, \lfloor N/2 \rfloor; \lfloor N/2 \rfloor + 1, \dots, N\}$
- Use $i \in S_1$ to estimate $\hat{M}(S_1)$ and formulate $\hat{H}_0(S_1)$
- Use $i \in S_2$ to calculate test-statistics to test $\hat{H}_0(S_1)$ under model specification $\hat{M}(S_1)$

Data splitting - Example

Remember, for our test to be valid in a classical sense, we must have

$$Prob\left(T \text{ rejected} | \hat{M}, \hat{H}_0\right) = Prob(T \text{ rejected})$$

- From *i.i.d.* (or ind.), X_i 's and ε_i 's are necessarily independent between these samples
- Hence, $\hat{M}(S_1) \perp\!\!\!\perp T(S_2)$, where $T(S_2)$ is the test statistic of interest estimated using S_2
- Also, any hypothesis formed from S_1 , $\hat{H}_0(S_1)$, is similarly independent

Data carving

Data “carving” is a very clever way to combine selective inference with data splitting

- [Fithian, Sun and Taylor (2014)]
- Goal: optimally balance the split used for model selection and for selective inference
- Outline:
 1. Use S_1 to select model
 2. Use $S_1 \cup S_2$ with selective inference
- The intuition of their solution is that conditioning on a random variable effectively disqualifies that variable for use as evidence against a hypothesis - hence want to condition on as little as possible in stage two
- This is beyond the scope of this course

Aside: Some comments and conjectures

- Classic inference in the context of post selection inference has been referred to as a “quiet scandal” (Leo Breiman, deceased Berkeley statistician)
- p -hacking is post-selection - but how can we possibly account for all the wrangling practised to obtain the desired p -values? We would need to know the full set of covariates the practitioner had at hand, then consider the full universe of combinations (polynomials, non-linear functions, etc.), THEN consider the subset of that universe that gives $p \ll 5\%$ for the desired regressor(s). Here is when workarounds like data splitting are useful
- IV can be post-selection: we only use instruments that are “strong” (mostly, there is a literature on weak instruments) - next lecture we look at inference after selection, but they take a different approach to truncated CDFs
- As mentioned, econometricians prefer not to make statements like $Y \sim N(\mu, \sigma^2 \mathbb{I}_{N \times N})$, but asymptotic theory without this assumption could be challenging in this setting.

References

- Fithian, W., D. Sun, and J. Taylor (2014). Optimal inference after model selection. *arXiv preprint arXiv:1410.2597*.
- Hastie, T., R. Tibshirani, and M. Wainwright (2015). *Statistical learning with sparsity: the lasso and generalizations*. CRC press.
- Lee, J. D., D. L. Sun, Y. Sun, and J. E. Taylor (2016). Exact post-selection inference, with application to the lasso.
- Tibshirani, R. J., J. Taylor, R. Lockhart, and R. Tibshirani (2016). Exact post-selection inference for sequential regression procedures. *Journal of the American Statistical Association* 111(514), 600–620.